USN | | | | | | | | | | [22CSE155]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

SENSOR TECHNOLOGIES

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

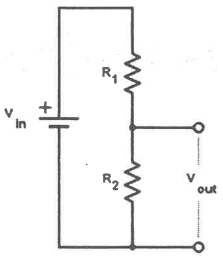
		PART-A	**5 X 2=10**		
1.	**a.**	Define Sensor. Give example for resistive sensor	L1	CO1	**02**
	b.	Mention the applications of LDR.	L2	CO2	**02**
	c.	Explain Self-inductance.	L2	CO3	**02**
	d.	List the types of different actuators.	L1	CO4	**02**
	e.	Mention the list of sensors used in Water monitoring system.	L2	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain Active transducer and its type.	L2	CO1	**05**
		OR			
	b.	Explain Dynamic and static errors.	L2	CO1	**05**
3.	**a.**	Define RTD? Explain its construction and operating principle. Why Platinum is used in RTD?	L3	CO2	**05**
		OR			
	b.	Mention the applications of thermistor	L3	CO2	**05**
4.	**a.**	Illustrate capacitive touch screen sensor and explain its basic elements.	L3	CO3	**05**
		OR			
	b.	Define Tachometer. Explain the working principle of the same.	L2	CO3	**05**
5.	**a.**	Illustrate pneumatic actuator and how does it work?	L3	CO4	**05**
		OR			
	b.	Explain Smart Material Actuators with an example.	L3	CO4	**05**
6.	**a.**	Explain Conveyor system with a neat diagram.	L2	CO5	**05**
		OR			
	b.	Explain weather monitoring system and mention the advantages	L2	CO5	**05**
7.	**a.**	Consider the following figure, 	L3	CO6	**05**

Fig 1

Fig.1 shows the potential divider circuit. Calculate the unknown quantity with each steps.

Sl.No	R_1	R_2	V_{in}	V_{out}
1	5KΩ	10KΩ	10V	?
2	?	10KΩ	5V	3V
3	7KΩ	?	10V	8V
4	8KΩ	2KΩ	?	10

		OR			
	b.	List out the advantages of the Transduction process.	L3	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Discuss the factors affects in measuring physical quantity	L2	CO1	**10**
		OR			
	b.	Explain Functional Units of Measuring system with neat sketch.	L3	CO1	**10**
9.	**a.**	Illustrate potentiometer. Mention its applications.	L3	CO2	**10**
		OR			
	b.	Explain LDR characteristics, construction, working principles.	L3	CO2	**10**
10.	**a.**	Explain Inductive Proximity Sensor. Mention its advantages and disadvantages.	L2	CO3	**10**
		OR			
	b.	Construct clap switch circuit with suitable components. Mention its applications.	L4	CO3	**10**
11.	**a.**	Explain thermal Actuator with neat diagram and mention benefits.	L4	CO4	**10**
		OR			
	b.	Illustrate Hydraulic Actuator. Explain its Working principle.	L3	CO4	**10**
12.	**a.**	Explain Wearable Technologies with an example.	L2	CO5	**10**
		OR			
	b.	Design health monitoring system and explain the components	L4	CO5	**10**
13.	**a.**	Consider an **"ELECTRONIC SYSTEM".** The expected value is 15 Voltage and measured value is 9 Voltage determine the **i.** Absolute error **ii.** Percentage of error **iii.** Accuracy **iv.** Relative accuracy	L4	CO6	**10**
		OR			
	b.	Design Home automation system using sensors and actuators.	L4	CO6	**10**